

- ① Literature is great.
You can include more experienced / residential segregation.
p.2.
- ② The empirical part is not clear.
The Entropy $\leftarrow$ gov facilitates mechanism needs more thinking.
p.5 p.6

How do sporting amenities affect social segregation

Economic complexity research proposal

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Introduction

Urban socio-economic inequalities keep increasing in cities, producing fragmented living environments where opportunities, services and social networks are unevenly distributed. Forces such as the location of residents' homes, the city's infrastructure and their movement in it shape people's opportunities and social contacts. In many cities the spatial configuration of neighbourhoods includes physical barriers and an unequal concentration of amenities, which reinforces socio-economic separation and limits interaction across groups. This plays a crucial role in the formation of segregation and in the fact that the divide between segregated communities grows stronger by the day if the factors and issues that create them aren't being handled correctly.

✓ cohesion

Budapest presents a clear case of socio-economic segregation, shaped by both historical legacies and post-socialist urban restructuring. Comparative research on European capitals shows that Budapest remains moderately segregated with a slightly increasing trend over time (Musterd et. al, 2016). With 23 districts and more than 1.6 million residents, Budapest is a big city filled with people from different socio-economic backgrounds and just a small walk could take you from one neighbourhood to another, that are drastically different from each other. Physical barriers are also present with a river parting the city and big, busy roads cutting it up into different and unique parts. Together, these factors show that segregation in Budapest is not necessarily incidental, but the outcome of historical infrastructure, natural and economic factors that continue to shape how different social groups occupy and experience the capital.

Pinker and
Lengel
2025

Amenities are an important factor in the fight against segregation, because they induce social mixing and diverse, well-distributed amenities can promote interaction that reduce network fragmentation. In our own experience this is especially true for a specific type of amenities: sporting facilities. According to us, sports can bring together residents from different backgrounds the most successfully, because not only do a lot of people share this hobby, but it is also common that you must work together with strangers you haven't met before, toward a shared goal, which doesn't really happen in other amenities. We have both met and created relationships with people from different backgrounds thanks to sports, which brings us to our question: Do sporting amenities have a positive effect on segregation?

← Atlas of
inequality

Some sports
are segregated
↓
football
↑
tennis

* Acad celebrities
Esteban Novo!
Marton Karsai
Zure Leskovec

Chetty

⇒ maybe distinguish
facilities by type?

In this research proposal, we will examine how different segregation is in the districts of Budapest, while considering the number of sporting amenities available for each district, and theorizing the correlation between the two.

Literature Review

Urban socio-economic segregation develops when groups with different income or educational backgrounds become unevenly distributed across neighbourhoods instead of mixing throughout the city (Van Ham et al., 2021). As inequalities have grown across European cities, these differences increasingly appear in the urban landscape, partly because higher-income households have more freedom to choose homes that match their preferences, which further strengthens the divide between advantaged and disadvantaged areas (Van Ham et al., 2021). The physical layout of cities also plays an important role: rivers, railways and the uneven location of amenities limit movement and interaction, making it harder for different social groups to meet in shared spaces and thereby reinforcing fragmented social environments (Tóth et al., 2021). Since segregation can also vary within administrative boundaries, recent research emphasises the importance of diversity-based measures such as entropy indices, which offer a much clearer picture of how mixed or separated neighbourhoods actually are (Östh et al., 2014). Comparative studies of European capitals further show that segregation has generally increased since the 2000s and is shaped by several larger forces at once: globalisation, inequality, and country-specific housing or welfare institutions, rather than a single cause (Tammaru et al., 2015). Together, these studies show that segregation is not simply an outcome of social differences, but a spatial process shaped by how urban structure channels movement, interaction, and everyday opportunities. This directly supports our research design: if segregation emerges partly from the uneven spatial organisation of cities, then examining how a specific type of amenity sports facilities is distributed across Budapest's districts can reveal whether they mitigate or reinforce these spatial divides.

Urban amenities matter because they structure where everyday interactions can take place. Oldenburg's (1999) well-known concept of "third places" highlights cafés, parks, recreational areas and other informal gathering spots as key locations where people meet others beyond their close social circles. Recent empirical work builds on this idea by showing how the spatial distribution of amenities shapes real patterns of social exposure. Chetty et al. (2022) demonstrate that cross-class interactions measured as the share of high-income friends among low-income individuals are strongly linked to higher levels of upward mobility across

neighbourhoods. Other research uses mobility data to show that popular activity hubs attract large and diverse groups of visitors, effectively determining where people have the chance to encounter one another (Schläpfer et al., 2021). Altogether, this literature highlights that amenities are not passive elements of the city, but active organisers of where interactions can realistically occur. Since both mobility and social mixing depend on where attractive hubs are located, investigating the spatial pattern of sports amenities becomes a relevant methodological step: if certain districts offer more accessible or more centrally positioned sports facilities, they may also create more opportunities for cross-group encounters, which can be captured by comparing district-level diversity metrics.

Studies focusing specifically on sports suggest that sports facilities are a unique form of social infrastructure. Longitudinal research shows that participation in community sport programmes regularly brings people together and helps build social capital, underlining the idea that sports settings can become routine sites of connection (Skinner, Zakus & Cowell, 2008). Spaaij's (2011) ethnographic work further demonstrates that recreational sport can create bonding, bridging and linking social capital across participants, offering direct opportunities for meaningful interaction between people from different backgrounds although these cross-group encounters do not always occur evenly. These findings imply that sports facilities can function as everyday meeting points where people regularly encounter others from different socio-economic backgrounds. For our research, this creates a clear theoretical mechanism: if sports spaces naturally encourage recurring interaction, then districts with more or more accessible facilities may show lower levels of socio-economic segregation. This justifies analysing sports amenities not just as infrastructure, but as potential "interaction generators" whose presence may influence district-level entropy values.

While the general literature shows how amenities shape social mixing, a more specific set of studies investigates how the distribution of sports facilities themselves reflects or reinforces inequality. Literature consistently finds that sports infrastructure is unevenly distributed across cities: wealthier or central neighbourhoods tend to have better access to high-quality facilities, while more peripheral or disadvantaged areas often face shortages. Shen (2020), for example, documents that in Nanning, most facilities are concentrated in the central area, leaving outer districts with significantly fewer opportunities for physical activity. Using detailed mobility and amenity data, Juhász et al. (2023) show that different types of amenities differ greatly in how socio-economically diverse their visitors are. Some attract people from many parts of the city, while others mainly draw homogeneous local groups. This means that the location and type of

amenities strongly influence whether everyday encounters between social groups are likely or remain rare. At the same time, research on regional sports participation shows that infrastructure alone does not always reduce inequalities: Kokolakis et al. (2014) find no significant effect of facility provision on participation rates across English Local Authorities, suggesting that social and demographic factors also matter. Taken together, these findings show that both the spatial distribution of sports facilities and the types of people they attract shape how different groups engage in sport and whether sports spaces ultimately act as bridges across socio-economic divides or simply reproduce existing inequalities. This reinforces the relevance of our district-level approach: by mapping how sports amenities are distributed across Budapest and comparing these patterns with socio-economic diversity measures, we can assess whether certain districts structurally enable more social mixing, or whether infrastructure disparities mirror and potentially deepen existing segregation patterns.

Data and method

To examine how the availability of sports amenities relates to social segregation across urban areas, we construct a district-level dataset for Budapest that integrates three independent data sources: population census data, geospatial information on sports facilities and socio-economic controls. We hope that the combination of these datasets allows us to capture both the social structure and the environment of each district.

The dependant variable is a segregation index, which is one we had to calculate ourselves from the data of the population census of 2022. We rely specifically on the distribution of education attainment categories, because there were no data on income, and we think this variable correlates the most with it. The 5 categories of education attainment are the following: less than primary education, primary education, secondary education without graduation, secondary education with graduation and higher education degree. Using these categories we calculated the entropy (James et. Al, 2013) for all the districts:

Equation 1: Entropy

$$Segregation_i = - \sum_{k=1}^N p_{ik} \times \log_2(p_{ik})$$

This helps us calculate how diverse or homogeneous a neighbourhood is, high entropy means high diversity and low segregation, while low entropy means the opposite. A limitation here is

!!!
Sports facilities
different in size
(a golf club cannot be in the 5th district)

Entropy may not capture the type of mixing you are pr.
Some sports facilities attract people from diff. districts, which is the major source of mixing.

that just because a diverse group of people live in one district, it doesn't mean they aren't segregated from each other in smaller neighbourhoods, but since the census data's smallest measuring format is districts, this was the closest we could get to quantifying segregation with the data available. The figures in Table 1 show how there is a significant difference between the segregation in the districts:

Table 1: The entropies of the districts of Budapest

District	Entropy	District	Entropy	District	Entropy
I.	1.72	IX.	1.95	XVII.	2.14
II.	1.77	X.	2.15	XVIII.	2.15
III.	2.04	XI.	1.86	XIX.	2.13
IV.	2.11	XII.	1.75	XX.	2.16
V.	1.82	XIII.	1.96	XXI.	2.15
VI.	1.90	XIV.	1.98	XXII.	2.12
VII.	1.99	XV.	2.14	XXIII.	2.17
VIII.	2.05	XVI.	2.09	-	-

For our independent variable we would use the number of sporting facilities in each district, which we would get from OpenStreetMap. Previous literature covers how not only the amenities in themselves are important, but their locations too, which would be interesting to examine in further research, but in this paper we will only concentrate on the number of facilities and hypothesize, that there is a connection between how many there are in a district, and how scattered they are in space. To reduce omitted variable bias, we could use district level demographic controls like population density, district area, etc.

We would use OLS regression to determine the connection between the variables using an equation like the following, probably with improvements, once we get to look at some results:

Equation 2: OLS regression

$$Segregation = \beta_0 + \beta_1 \times Sport_facilities + \sum \beta_k \times Control_variable$$

This equation does not allow for the type of amenity role argued in the motivation.

The link is not clear: ~~sport facilities~~ what can ~~the~~ the cross-section tell us?

Expected outcomes, impact, and limitations

Based on our hypothesis and the patterns we see in the data so far, we expect to find connection between the number of sports facilities in a district and the level of socio-economic segregation. More precisely, we think that districts with more sports amenities will show higher entropy values, meaning more diverse populations and lower segregation. The reasoning behind this is that sports facilities can serve as shared spaces where people from different backgrounds may interact in their daily routines, which could make districts more socially mixed over time. Even though our analysis cannot directly measure social interaction, a positive relationship between facilities and diversity would be an important first indication that sports amenities might play a role in reducing social separation.

If our results support this, the study could have useful implications. On the academic side, it would contribute to the growing literature that connects the spatial distribution of urban amenities with segregation patterns. Sports facilities are especially interesting because they are cheaper and more accessible than many other amenities, so their potential impact on social mixing could be stronger. On the policy side, if more sports facilities correlate with less segregation, this could support strategies where cities invest more in recreational infrastructure in districts that are more socially homogeneous.

At the same time, our study has several limitations. One important issue is that the census data is only available at the district level, which is quite a large unit. Within-district differences can be very big, so our segregation index may hide smaller neighbourhood-level patterns. Another limitation is that we only look at the number of sports facilities, not their quality, size, price, or how easy they are to reach by foot or public transport. These factors could strongly influence who uses them. We also cannot observe how crowded these facilities are, or whether they are used mostly by locals or by people from other districts. There are also several omitted variables that might affect both segregation and sports amenities. For example, housing prices, local labour market conditions, access to schools, crime rates, and even individual variables that are immeasurable (e.g., diligence, health, etc.) could all influence how diverse a district is and how many facilities it should have. These factors are difficult to measure, but future versions of the study could incorporate them to reduce omitted variable bias.

In terms of improving the methodology, as we mentioned before the next step could be to move from district-level data to smaller spatial units, such as census tracts or $1 \text{ km} \times 1 \text{ km}$ grid cells. This would give us a more accurate picture of how segregation appears across the city. Another

Yes, but
for that
you need
dynamic
data

segregated
districts can
have more
sport clubs
to separate
people

rich gym
poor gym

potential improvement would be to include the spatial distribution of sports facilities within each district. Instead of counting how many facilities exist, we could calculate how close the average resident is to a sports amenity, or how evenly facilities are spread out. Additionally, if mobility data (e.g., Google mobility, telecom data) becomes available, we could examine actual usage and not just availability. Finally, the study could be extended by using other methods such as spatial regression models, which can capture the relationships between neighbouring districts, or even instrumental variables if a valid instrument for sports facility placement could be identified. These approaches would allow us to better estimate the causal effect of sports amenities on segregation. Overall, even with its limitations, our research can provide a useful starting point for understanding whether the availability of sports facilities plays a role in shaping the social structure of Budapest's districts, and it opens several directions for more detailed future work.

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alphabetical order ☺

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from this
add more
on experienced
segregation
MIT group includes
More
Portland

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